[Spring 2024] Deep Learning - Final-Project
Building Use Type Classification in Dense Urban Environments using Deep Learning and Aerial Imagery

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GitHub Repository: https://github.com/matzefit/AIBuildingFunctionPredicition

Abstract

This study employs a U-Net architecture to classify building functions in dense urban environments using high-resolution aerial imagery. The model effectively segments multiple building types but faces challenges distinguishing between similar classes and separating buildings from dense vegetation. Despite these limitations, the present study can be seen as a pilot project, when further refined, offering valuable insights into the urban layout and building stock of Cities where building functions aren't extensively mapped or made openly accessible.

Introduction

Automatization of land cover and land use classification tasks are established research topics among the researchers and practitioners in the fields of geography and GIS, remote sensing, planning and agriculture among many others (ScienceDirect 2024). With the abundance of low cost and often high resolution satellite and aerial imagery and increased usage of Deep Learning models, this traditionally mundane task can now be achieved in a matter of seconds. There are lots of techniques and tools to successfully identify various types of land cover (Talukdar et al. 2020). However, not many studies exist that try to identify the specific function of buildings, especially in the dense and complex urban landscape. Among the few studies dealing with building classifications in dense cities, Huang et al. (Huang et al. 2022) introduced a dataset covering the cities of Beijing and Munich, enabling detailed classification of building roof types using high-resolution satellite images. Dimassi et al. (Dimassi et al. 2021) introduced a transfer learning approach to classify building damage and building types, utilizing high-resolution satellite images. Their model leverages a pre-trained ResNet50 on ImageNet and subsequently applies Mask R-CNN for instance segmentation. In contrast, our study aims to develop a lean model for the semantic classification of building use types without relying on transfer learning. As an additional output of this study we provide a publicly available training dataset with an equally high granularity as in (Huang et al. 2022) but for building functions.

Methodology

Training Data Preparation

The training data for this project consists of two processed datasets: true color aerial imagery and mask of the building footprints mapped to the corresponding land use as a label.

a) Aerial Imagery: National Agriculture Imagery Program (NAIP) The National Agriculture Imagery Program (NAIP) features recent high-resolution aerial imagery from the USDA Farm Services Agency. NAIP imagery is collected annually during growing season in the continental United States. The NAIP imagery is publicly available and is released within a year of collection. Since 2018, the images resolution is standardized across the regions and equals 0.6 meters per pixel, making it one of the best quality satelite imagery data publicly and freely available data in the US. For the purpose of this study we used images of New York City from 2022 (U.S. Geological Survey 2022). For the sake of computation feasibility we downscaled the resolution of the images to 1.953125 meters per pixel resulting in 128 pixels per 250 m.

b) Land Use Mask To generate the land use mask used as the ground truth for the training dataset we relied on two spatial datasets available in New York City. First is the building footprint data that provides precise location of each structure in the City and will serve as a mask to locate each property(City of New York 2020a). Building land use type information used as a proxy of building function, was obtained from publicly available land use data called Primary Land Use Tax Lot Output (PLUTO) provided by NYC Dept. of City Planing(City of New York 2020b). It is reported on the tax lot level. To assign each property with the land use information we performed spatial join operation using Python and Geopandas environment. The resulting shapefile was then rasterized, in accordance to the data describing the aerial imagery using GDAL framework. Finally, both aligned rasters were sliced into 250 m tiles represented by 128 pixels. Tiles not representing any property were dropped.

The combined training data processing, resulted in a total of 19,315 image-mask pairs. The most ideal data split has been identified as 70% of the data used for training (resulting

Figure 1: Rasterized NAIP Aerial Imagery (left) and land use data(right)

in 13520 images), and 15% each for validation and testing (2897 and 2898 images respectively).

It can be argued that the entirety of the data describes only the area of New York City. This introduces a certain bias towards the classification fitting only to a certain range of architectures and city layout specific to NYC. We considered therefore the test of model predictions in other areas, however this posed to be challenging as other city administrations integrate different function typologies than in NYC making a verification of results difficult. The scope of this project is therefore limited to the area of NYC only.

The building functions included in the training data, as illustrated in Figure 2, reveal a significant class imbalance, with the 'Background' class being overwhelmingly dominant, followed by 'One & Two Family Buildings', and a much smaller representation of other building functions. This imbalance of building functions is typical for urban datasets which is why in our current implementation, we did not employ specific measures to mitigate class imbalance, such as class weighting or data augmentation techniques targeted at underrepresented classes. Consequently, the trained model's output inherently reflects these imbalances. While this approach maintains the integrity of the dataset's representation, it may impact the model's performance on less frequent classes. Future work could explore strategies to address this imbalance to enhance the model's ability to accurately predict underrepresented building functions, potentially improving the overall performance and robustness of the segmentation predictions.

Model Details

The model's task is to segment images containing multiple buildings across different scales, and to distinguish among various building functions in a semantic segmentation context. U-Net architectures, widely acclaimed for their effectiveness in semantic segmentation tasks (Ronneberger, Fischer, and Brox 2015), have been chosen for this purpose due to their suitability for semantic segmentation through pixel-level classification rather than instance segmentation aiming at the differentiation of individual objects in an image. Although vision transformer models are powerful for capturing global dependencies, they present greater complexity and computational demands. These factors do not align with our project's resource constraints and timeframe. Additionally, vision transformers typically require larger datasets to perform effectively, which poses a challenge given our focused dataset size. These reasons solidify our choice of sticking with a U-Net based architecture. The resulting model architecture can be seen in figure 3. The model consists of 17268108 trainable parameters and has been trained for 35 epochs that took ca. 75 mins to complete on a NVIDIA Geforce RTX 4070 Graphics card.

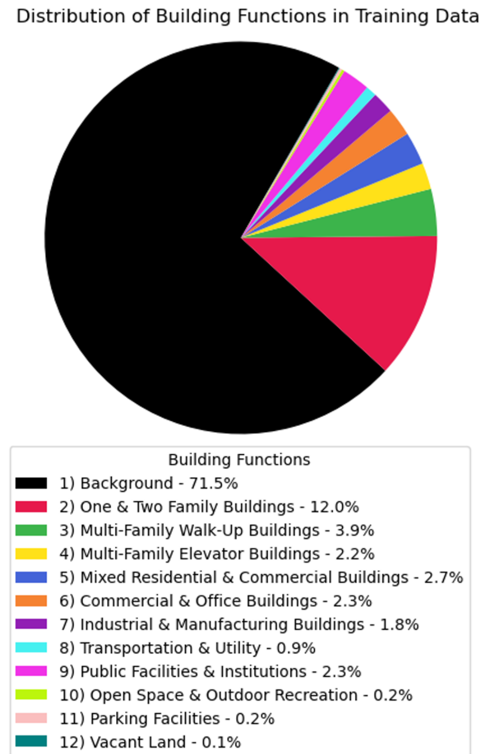

Figure 2: Distribution of building functions/Land Use in the training data

Hyperparameter Tuning and Architecture Tests

Various tests with different selections of hyperparameters were conducted. The most optimal setup was identified as using AdamW with a weight decay of $1e-03$, a cyclic learning rate scheduler in triangular mode, and a base learning rate starting at $1e-05$ ramping up to $1e-03$ within 5 epochs. No gradient clipping was applied, as it was found to deteriorate validation accuracy. In addition to hyperparameter tests, a range of modifications to the U-Net architecture were explored. Introducing extensive dilated convolutions in the encoder part of the U-mesh resulted in less sharp representation of building edges in the predicted mask. However, limiting the application of dilated convolutions to only the last two downsampling blocks on the encoder side of the network increased the accuracy of correctly classified pixels in the test dataset. This improvement likely results from the enhanced ability of dilated convolutions to capture broader contextual information without losing spatial resolution. Experiments were conducted to determine if incorporating additional ResNet-like skip connections within the double convolutions in both the encoder and decoder would improve the observed loss function pattern. This modification led to the cross-entropy loss for validation deviating much earlier during training. However, the IoU value for the validation data showed significant improvement, often reaching around 0.74 in epochs ≤ 10. Despite this improvement, the model with additional skip connections within the double convolution blocks exhibited overfitting when evaluated on test data, with IoU scores around 0.68 compared to the current model's IoU score of 0.71. Further experiments with SqueezeExcitation blocks and self-attention surprisingly produced worse results than the current implementation of the U-Net. Experiments with a different resolution and tile size within the training data, specifying that 128 pixels cover 250m rather than 500m, resulted in a substantial increase in accuracy (IoU increased from 0.54 to 0.71).

Results

The predictions on the test dataset, as shown in Figure 4, reveal several limitations of the model. One significant issue is the misclassification of buildings into similar classes. This is evident in the confusion matrix (Figure 5), where there is notable confusion between "2) One & Two Family Buildings" and "3) Multi-Family Walk-Up Buildings". Another challenge is the model's difficulty in distinguishing between "11) Parking Facilities," "6) Commercial & Office Buildings," and "7) Industrial & Manufacturing Buildings". These building functions appear very similar in airborne imagery, making it difficult for the model to identify distinct features and accurately classify them.

A significant challenge lies in distinguishing between building functions and background pixels, which constitute 71% of all labels, compared to the "12) Vacant Land" class, which comprises only 0.1%. In many instances, the aerial images contain substantial vegetation, complicating the identification of buildings. The "Vacant Land" class is particularly problematic, as it is frequently misclassified as background due to buildings with this function often being obscured by dense vegetation canopies, further hindering accurate classification.

Another problem is that the model has in certain cases difficulties to identify distinct building outlines, none of the mentioned alternatives in the architecture setups has been able to improve on the edge detection. It is likely that using a much deeper network, a modified loss function, or the pre-application of edge highlighting techniques could enhance the geometrical quality of the detected building outlines.

On a positive note, overfitting has been relatively well mitigated through network optimization and hyperparameter tuning. As reflected in Figure 6, the validation loss does not lie significantly above the training loss. The maximum validation IoU score of 0.7123 visible in 7 was achieved at epoch 21, indicating that substantial parameter tuning has effectively maximized classification accuracy.

The results reveals intriguing patterns in the classifications. Notably, it consistently assigns a uniform class to buildings located near major streets in densely built areas. These buildings are primarily classified as manufacturing and industrial buildings, though they should more accurately be classified as commercial buildings. This uniform assignment suggests that, if properly trained, the model could provide deeper insights into the underlying rules and patterns governing the urban fabric and building stock in US cities. Such insights could be valuable for urban planning and development, highlighting distinctive characteristics of urban environments.

Conclusion

The present model is capable of mapping building functions, solely through RGB aerial images. The procedure is therefore a promising approach for areas where building functions are not publicly provided, which in some research fields related to GIS (Geoinformationsystems) and urban science can be a crucial information. Despite challenges in distinguishing similar classes and accurately identifying buildings in vegetated areas, the model shows potential for offering valuable insights into urban layouts and building functions. Overfitting has been effectively mitigated through network optimization and hyperparameter tuning. Future improvements could be achieved with a deeper architecture, a modified loss function, and even more extensive training data. These enhancements could further refine the model's accuracy and applicability, making it a robust tool for urban planning and development.

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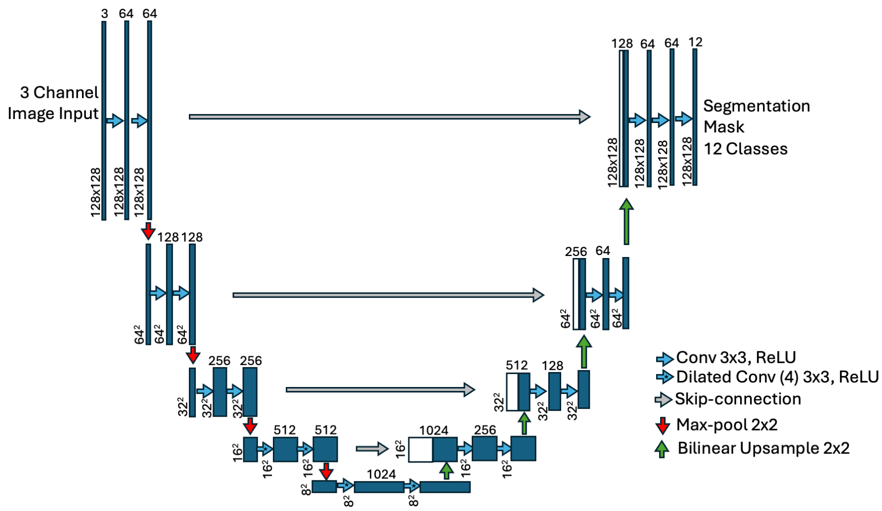

Figure 3: U-Net Architecture with Dilated Convolutions used in this project

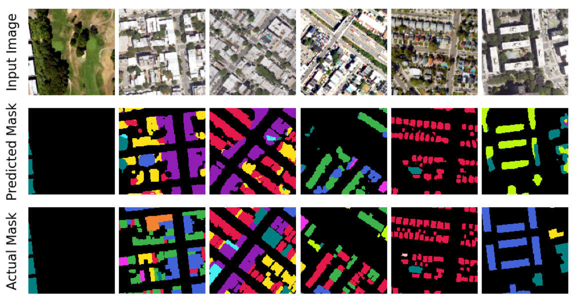

Figure 4: Predictions on six random images of the test dataset, the coloring of the building functions follows legend of figure 2.

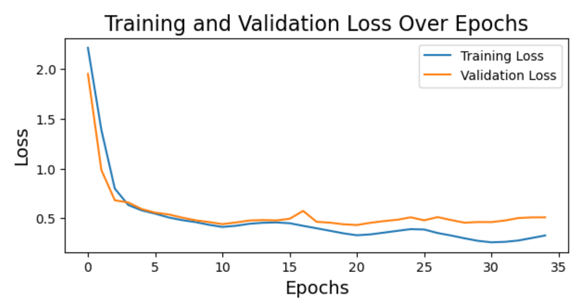

Figure 6: Training and Validation Loss over Epochs

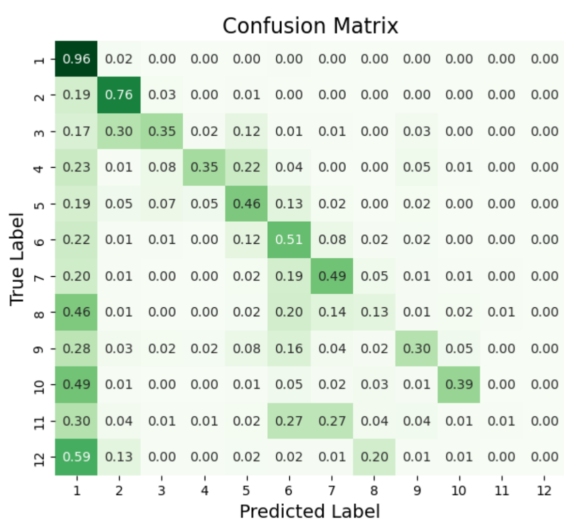

Figure 5: Confusion Matrix for predicting pixels per building function, Numbering of labels follows legend of figure 2

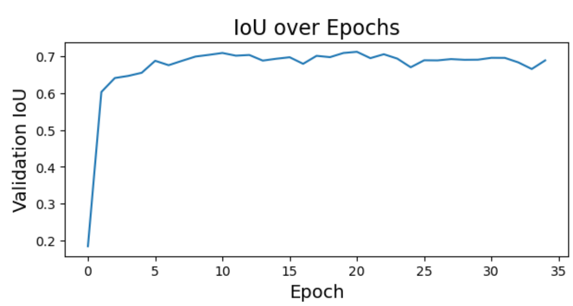

Figure 7: Intersection over Union per Epoch. Best result on validation data in epoch 21 with 0.7123